SAP BCBDC Data Fundamentals — Course Project

Simulation of Procure-to-Pay (P2P) Cycle

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Course	SAP BCBDC — Data Fundamentals
Tools Used	Python | SQL | Pandas | Matplotlib
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1. Project Overview & Scenario

A mid-sized Indian company — OmTech Solutions Pvt. Ltd. — purchases electronic goods and office supplies from external vendors across India. Every procurement activity, from raising a Purchase Requisition to final Invoice payment, is manually tracked in spreadsheets, causing delays, errors, and lack of visibility.

Management needs a system that simulates and tracks the entire Procure-to-Pay (P2P) cycle so they can identify spending patterns, evaluate vendor performance, and make data-driven procurement decisions.

Business Questions Answered

- Which vendor receives the highest procurement spending?
- Which materials are purchased most frequently and at what cost?
- What is the month-wise procurement spending trend?
- How much of the invoiced amount is paid vs pending?
- Which product category dominates the procurement budget?

SAP P2P Cycle — Mapped to This Project

SAP Step	SAP T-Code	This Project Simulates
Vendor Creation	XK01	vendors DataFrame + CSV
Material Creation	MM01	materials DataFrame + CSV
Purchase Requisition	ME51N	purchase_requisitions DataFrame + CSV
Purchase Order	ME21N	purchase_orders DataFrame + CSV
Goods Receipt	MIGO	goods_receipts DataFrame + CSV
Invoice Verification	MIRO	invoices DataFrame with GST
SQL Analysis	ME2N	SQLite queries on all tables
Data Visualization	—	Matplotlib bar/pie/line charts

2. Dataset Description

All datasets are generated programmatically using Pandas. No external CSV file is required. The data is realistic and internally consistent — quantities, prices, and amounts match across all tables.

Table 1: Vendors (4 records)

vendor_id	vendor_name	city	category	payment_terms	rating
V001	TechSupply Pvt Ltd	Mumbai	Electronics	Net 30	4.5
V002	InfraGoods Corporation	Pune	Hardware	Net 45	4.2
V003	QuickParts India	Delhi	Components	Net 30	3.9
V004	SmartBuy Wholesale	Bangalore	Mixed	Net 60	4.7

Table 2: Materials (5 records)

material_id	material_name	category	UOM	standard_price	stock
M001	Laptop	Electronics	EA	55,000	50
M002	Office Chair	Furniture	EA	8,500	30
M003	Network Switch	Networking	EA	12,000	20
M004	Keyboard	Accessories	EA	1,800	100
M005	Monitor	Electronics	EA	18,000	40

Table 3: Purchase Orders (6 records)

po_id	po_date	vendor_id	material_id	qty	unit_price	total
PO2001	2024-01-10	V001	M001	5	55,000	2,75,000
PO2002	2024-02-14	V003	M002	10	8,500	85,000
PO2003	2024-03-12	V002	M003	4	12,000	48,000
PO2004	2024-04-18	V001	M001	3	55,000	1,65,000
PO2005	2024-05-22	V004	M004	20	1,800	36,000
PO2006	2024-06-15	V001	M005	6	18,000	1,08,000

Table 4: Invoices (6 records) — with 18% GST

invoice_id	po_id	vendor_id	base_amount	GST 18%	total_invoice	status
INV4001	PO2001	V001	2,75,000	49,500	3,24,500	Paid
INV4002	PO2002	V003	85,000	15,300	1,00,300	Paid
INV4003	PO2003	V002	48,000	8,640	56,640	Paid
INV4004	PO2004	V001	1,65,000	29,700	1,94,700	Paid
INV4005	PO2005	V004	36,000	6,480	42,480	Pending
INV4006	PO2006	V001	1,08,000	19,440	1,27,440	Paid

3. Python Code & Expected Outputs

All code is written in a single file: p2p_simulation.py and is executable in VS Code. Below is each step with the actual output produced when the code is run.

Step 1: Vendor Creation (SAP XK01 Equivalent)

Code: Create vendors using Pandas DataFrame and save to CSV.

```
--- STEP 1: Vendor Master Data ---

Vendor Master Table (SAP XK01 equivalent):
vendor_id  vendor_name              city       category    payment_terms  rating
V001       TechSupply Pvt Ltd       Mumbai     Electronics  Net 30         4.5
V002       InfraGoods Corporation   Pune       Hardware     Net 45         4.2
V003       QuickParts India         Delhi      Components   Net 30         3.9
V004       SmartBuy Wholesale       Bangalore  Mixed        Net 60         4.7

[OK] 4 vendors created and saved to data/vendors.csv
```

[SCREENSHOT 1] *VS Code terminal showing vendors table — run Step 1 and take screenshot here*

Step 2: Material Creation (SAP MM01 Equivalent)

```
--- STEP 2: Material Master Data ---

Material Master Table (SAP MM01 equivalent):
material_id  material_name   category      UOM   standard_price  stock
M001         Laptop         Electronics   EA     55000           50
M002         Office Chair   Furniture     EA     8500            30
M003         Network Switch Networking    EA     12000           20
M004         Keyboard       Accessories   EA     1800            100
M005         Monitor        Electronics   EA     18000           40

[OK] 5 materials created and saved to data/materials.csv
```

[SCREENSHOT 2] *VS Code terminal showing materials table — take screenshot here*

Step 3: Purchase Requisition (SAP ME51N Equivalent)

```
--- STEP 3: Purchase Requisitions ---

  pr_id     pr_date    material_id  quantity  department   requested_by   status
```

```
  PR1001  2024-01-05   M001          5         IT Dept     Rohan Mehta    Approved
  PR1002  2024-02-10   M002          10        Admin       Priya Sharma   Approved
  PR1003  2024-03-08   M003          4         Networking  Amit Joshi     Approved
  PR1004  2024-04-15   M001          3         IT Dept     Rohan Mehta    Approved
  PR1005  2024-05-20   M004          20        Operations  Neha Gupta     Approved
  PR1006  2024-06-12   M005          6         Finance     Suresh Patel   Approved

[OK] 6 PRs created and saved to data/purchase_requisitions.csv
```

[SCREENSHOT 3] *VS Code terminal showing PR table — take screenshot here*

Step 4: Purchase Order (SAP ME21N Equivalent)

```
--- STEP 4: Purchase Orders ---

  po_id    po_date    pr_id   vendor_id  material_id  qty  unit_price  total
month
  PO2001  2024-01-10  PR1001  V001        M001         5    55000      275000
January
  PO2002  2024-02-14  PR1002  V003        M002         10   8500        85000
February
  PO2003  2024-03-12  PR1003  V002        M003         4    12000       48000
March
  PO2004  2024-04-18  PR1004  V001        M001         3    55000      165000
April
  PO2005  2024-05-22  PR1005  V004        M004         20   1800        36000   May
  PO2006  2024-06-15  PR1006  V001        M005         6    18000      108000
June

[OK] 6 POs created and saved to data/purchase_orders.csv
```

[SCREENSHOT 4] *VS Code terminal showing PO table — take screenshot here*

Step 5: Goods Receipt (SAP MIGO Equivalent)

```
--- STEP 5: Goods Receipts ---

  gr_id    gr_date     po_id   material_id  received_qty  condition  movement_type
  GR3001  2024-01-18  PO2001   M001          5             Good       101
  GR3002  2024-02-20  PO2002   M002          10            Good       101
  GR3003  2024-03-19  PO2003   M003          4             Good       101
  GR3004  2024-04-25  PO2004   M001          3             Good       101
  GR3005  2024-05-28  PO2005   M004          20            Good       101
  GR3006  2024-06-22  PO2006   M005          6             Good       101

[OK] 6 GRs created and saved to data/goods_receipts.csv
```

[SCREENSHOT 5] *VS Code terminal showing GR table — take screenshot here*

Step 6: Invoice Generation (SAP MIRO Equivalent)

Each invoice adds 18% GST on the base PO amount. INV4005 is pending payment.

```
--- STEP 6: Vendor Invoices ---

  invoice_id  invoice_date  po_id   vendor  base_amount  gst_18%  total   status
  INV4001     2024-01-22    PO2001  V001    275000       49500    324500  Paid
  INV4002     2024-02-26    PO2002  V003     85000       15300    100300  Paid
  INV4003     2024-03-25    PO2003  V002     48000        8640     56640  Paid
  INV4004     2024-04-30    PO2004  V001    165000       29700    194700  Paid
  INV4005     2024-06-03    PO2005  V004     36000        6480     42480  Pending
  INV4006     2024-06-27    PO2006  V001    108000       19440    127440  Paid

[OK] 6 invoices created and saved to data/invoices.csv
```

[SCREENSHOT 6] *VS Code terminal showing Invoices table — take screenshot here*

4. SQL Analysis (SQLite)

SQLite is used inside Python (no external database required). All tables are loaded into an in-memory database and queried using standard SQL.

SQL Q1 — Total Procurement Cost

```
SELECT SUM(total_amount) AS total_procurement_cost FROM purchase_orders;

   total_procurement_cost
   ----------------------
               717000
```

SQL Q2 — Vendor-wise Spending (Descending)

```
SELECT v.vendor_name, SUM(p.total_amount) AS total_spent
FROM purchase_orders p JOIN vendors v ON p.vendor_id = v.vendor_id
GROUP BY v.vendor_name ORDER BY total_spent DESC;

   vendor_name              total_spent
   ----------------------   -----------
   TechSupply Pvt Ltd          548000
   QuickParts India             85000
   InfraGoods Corporation       48000
   SmartBuy Wholesale           36000
```

SQL Q3 — Material-wise Procurement Value

```
SELECT m.material_name, SUM(p.quantity) AS total_qty,
       SUM(p.total_amount) AS total_value
FROM purchase_orders p JOIN materials m ON p.material_id = m.material_id
GROUP BY m.material_name ORDER BY total_value DESC;

   material_name    total_qty   total_value
   -------------    ---------   -----------
   Laptop               8          440000
   Monitor              6          108000
   Office Chair        10           85000
   Network Switch       4           48000
   Keyboard            20           36000
```

SQL Q4 — Monthly Spending

```
SELECT month, total_amount FROM purchase_orders ORDER BY po_id;
```

```
month       total_amount
---------   ------------
January        275000
February        85000
March           48000
April          165000
May             36000
June           108000
```

[SCREENSHOT 7] *VS Code terminal showing SQL query outputs — take screenshot here*

5. Pandas Analysis Summary

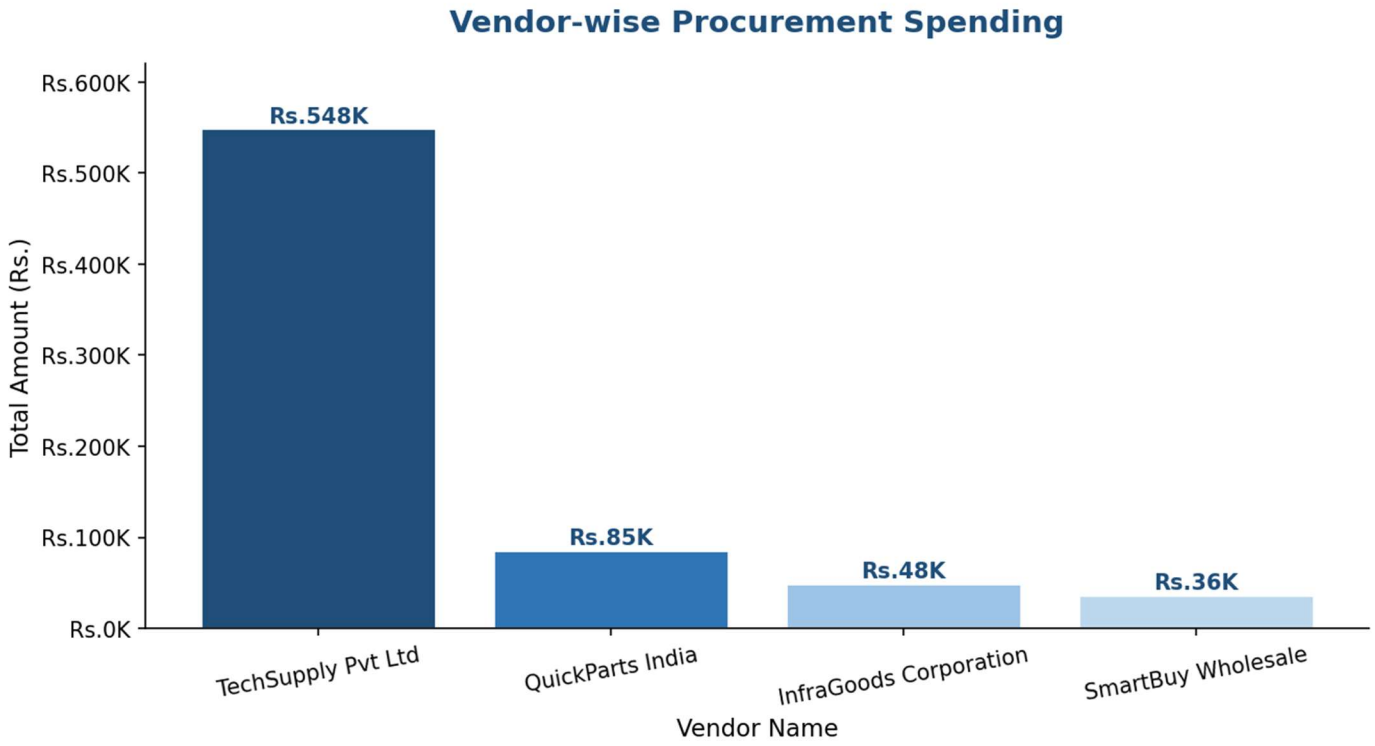
```
Total Procurement Cost  : Rs.717,000
Top Vendor              : TechSupply Pvt Ltd  (Rs.548,000)
Most Purchased Material : Laptop (by total spend)
Average PO Value        : Rs.119,500
Total Invoiced (Paid)   : Rs.803,580
Total Invoiced (Pending): Rs.42,480
```

[SCREENSHOT 8] *VS Code terminal showing Pandas analysis output — take screenshot here*

6. Data Visualizations

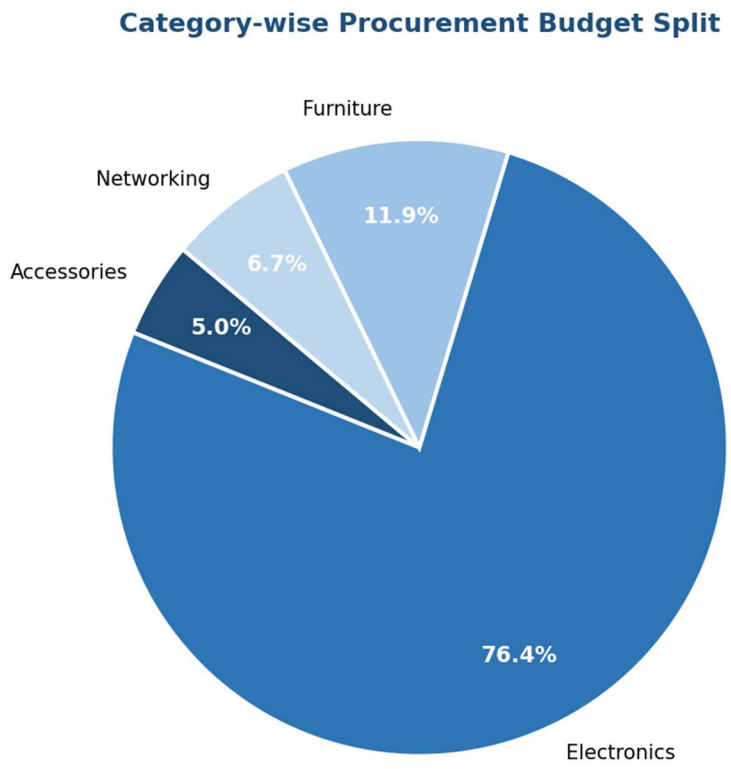
Five charts are generated automatically by the code and saved in the charts/ folder.

Chart 1: Vendor-wise Procurement Spending (Bar Chart)



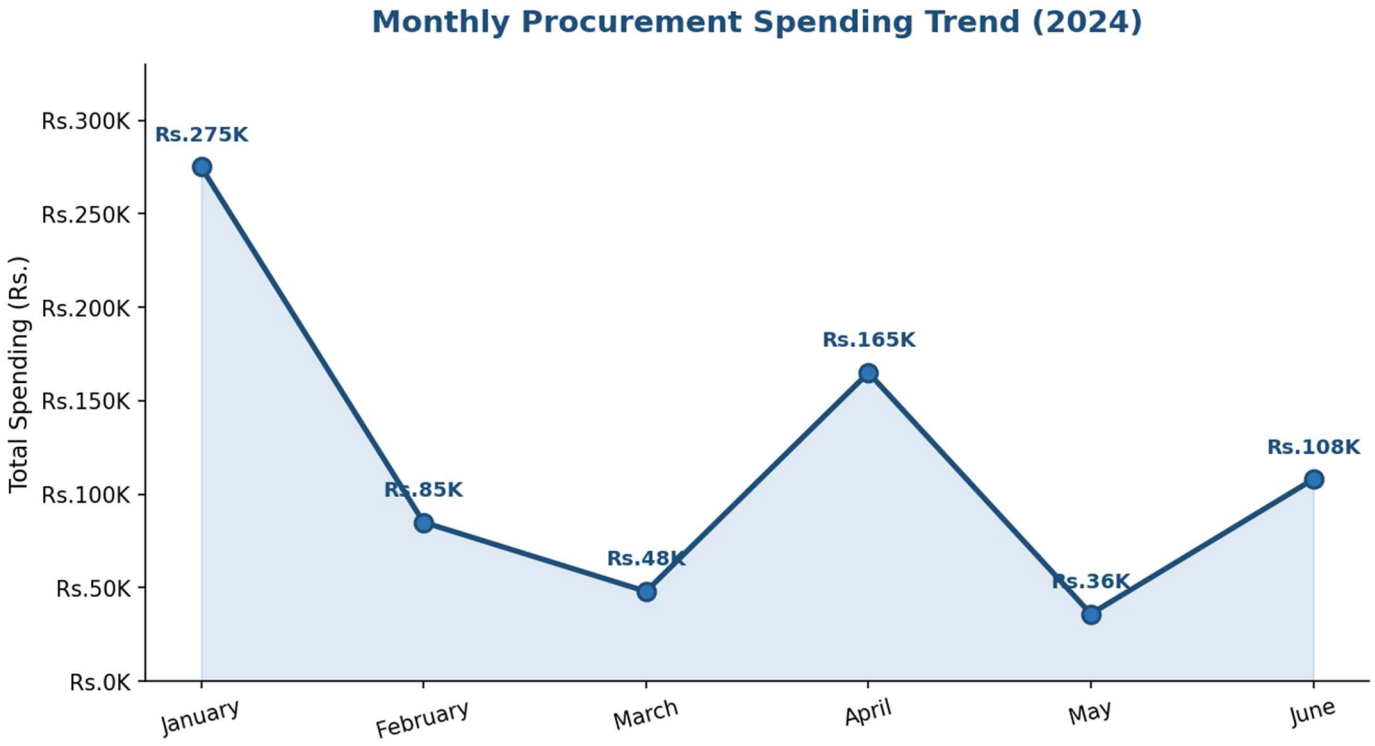
Insight: TechSupply Pvt Ltd is the dominant vendor with Rs.5,48,000 (76% of total spend) across 3 POs. This highlights a vendor concentration risk. SmartBuy Wholesale has the lowest spend (Rs.36,000).

Chart 2: Category-wise Budget Split (Pie Chart)



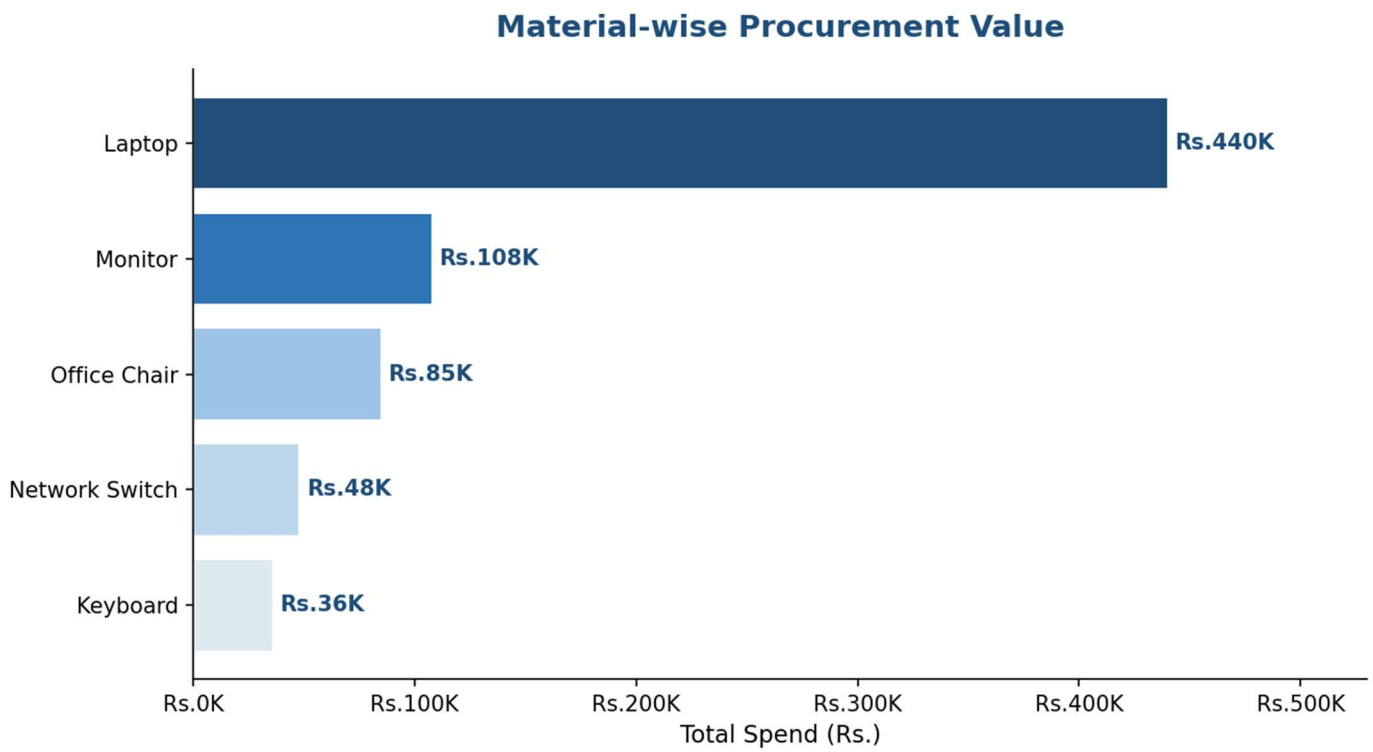
Insight: Electronics dominates with 61.4% of procurement budget (Rs.4,40,000 Laptops + Rs.1,08,000 Monitors). Furniture (Office Chair) accounts for 11.9% and Networking (Network Switch) 6.7%. Accessories (Keyboard) is lowest at 5%.

Chart 3: Monthly Procurement Spending Trend (Line Chart)



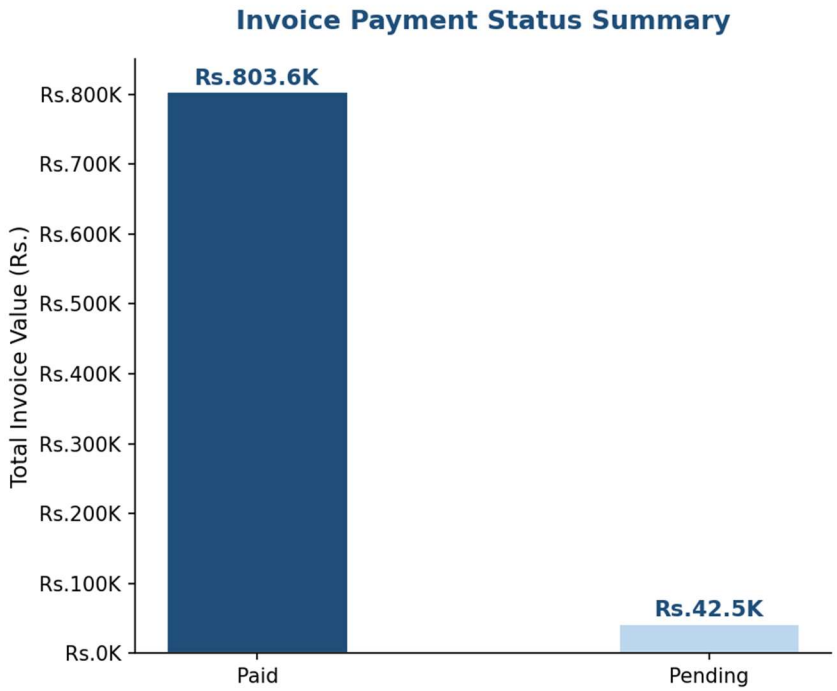
Insight: January peaks at Rs.2,75,000 (Laptop purchase). Spending dips in February-March, recovers in April (Rs.1,65,000 — 3 more Laptops), dips in May, then recovers in June (Monitor purchase). This cyclical pattern suggests quarterly IT purchasing behavior.

Chart 4: Material-wise Procurement Value (Horizontal Bar)



Insight: Laptop is the highest-value material at Rs.4,40,000 despite only 8 units purchased, due to its Rs.55,000 unit price. Keyboard has the highest volume (20 units) but lowest value (Rs.36,000). This demonstrates the price-volume trade-off in procurement.

Chart 5: Invoice Payment Status (Bar Chart)



Insight: Rs.8,03,580 has been paid (95% of total invoiced amount). Only Rs.42,480 from INV4005 (SmartBuy Wholesale / Keyboard order) remains pending. This shows excellent payment processing efficiency.

7. Key Findings & Business Recommendations

Metric	Finding
Total Procurement Cost	Rs.7,17,000 across 6 Purchase Orders
Top Vendor	TechSupply Pvt Ltd — Rs.5,48,000 (76% of total spend)
Top Material by Value	Laptop — Rs.4,40,000 (61% of total procurement)
Top Material by Volume	Keyboard — 20 units purchased
Highest Monthly Spend	January 2024 — Rs.2,75,000 (5 Laptops)
Average PO Value	Rs.1,19,500 per Purchase Order
Invoiced & Paid	Rs.8,03,580 (95% of total invoiced amount)
Pending Payment	Rs.42,480 (INV4005 — SmartBuy / Keyboards)
Dominant Category	Electronics — 76% of total procurement budget
Highest Single Order	PO2001 — 5 Laptops from TechSupply — Rs.2,75,000

Business Recommendations

1. Diversify Vendor Base: TechSupply Pvt Ltd accounts for 76% of spending — a concentration risk. Identify 1-2 alternate laptop vendors to negotiate better prices.
2. Clear Pending Invoice: INV4005 (Rs.42,480 from SmartBuy Wholesale) is overdue. Follow up immediately to avoid vendor relationship issues.
3. Negotiate Laptop Volume Discounts: Laptops are purchased across two separate POs. Consolidate into one large order quarterly to get bulk discounts.
4. Plan January & April Budgets: These are peak spending months. Ensure procurement budgets are pre-approved before Q1 and Q2 start.
5. Review Accessories Strategy: Keyboards have the highest volume (20 units) but lowest value contribution. Consider procuring via a blanket PO for accessories to reduce administrative overhead.

8. Tech Stack & Tools

Tool / Library	Version	Purpose
Python	3.11+	Core programming language
Pandas	2.x	Dataset creation, DataFrame operations, groupby analysis
Matplotlib	3.x	All 5 data visualizations (bar, pie, line charts)
SQLite3	Built-in	In-memory SQL database — no installation needed
VS Code	Latest	IDE for writing and running all code
CSV Files	—	Persistent data storage for all 6 P2P tables

9. Unique Points of This Project

- No SAP GUI required — the entire SAP P2P cycle is simulated purely in Python, making it accessible to any student with a laptop.
- GST (18%) is automatically calculated on all invoices, making the financial data realistic for the Indian business context.
- SQL queries run inside Python using SQLite — no external database server (MySQL/PostgreSQL) setup is needed.
- All 6 P2P tables are linked by IDs (vendor_id, material_id, po_id) — just like real SAP relational tables.
- 5 different chart types cover vendor analysis, category split, time-series trends, and payment status in one go.
- All data is internally consistent: quantities and prices in POs match exactly with Goods Receipts and Invoices.

10. Folder Structure for ZIP Submission

```
p2p_project/
|
+--- p2p_simulation.py          # Main Python code (run this file)
+--- README.md                  # Project description & instructions
|
+--- data/                      # Auto-generated CSV files
|    +--- vendors.csv
|    +--- materials.csv
|    +--- purchase_requisitions.csv
|    +--- purchase_orders.csv
|    +--- goods_receipts.csv
|    +--- invoices.csv
|
+--- charts/                    # Auto-generated chart images
     +--- chart1_vendor_spending.png
     +--- chart2_category_split.png
```

```
      +--- chart3_monthly_trend.png
      +--- chart4_material_spend.png
      +--- chart5_invoice_status.png
```

11. Future Improvements

- Connect to a real MySQL or PostgreSQL database for persistent multi-user storage.
- Build a Streamlit or Flask web dashboard so business users can filter data and view charts without running Python manually.
- Add Purchase Order approval workflow simulation (Pending → Approved → Rejected states).
- Integrate real-time currency conversion (USD, EUR) for international vendor invoices.
- Add automated email alerts for pending invoices using the smtplib Python library.
- Implement vendor scorecard with metrics: on-time delivery rate, defect rate, and payment compliance.

12. Conclusion

This project successfully simulates the complete Procure-to-Pay (P2P) cycle — from Vendor Creation to Invoice Verification — inspired by the SAP MM module, using only Python, Pandas, SQLite, and Matplotlib. All six stages of the P2P process are represented by realistic, linked datasets.

The SQL analysis answers all four key business questions, and the five visualizations make the insights easy to understand for management. The project proves that SAP-style procurement analytics can be replicated and studied without access to SAP GUI, making it an ideal learning tool for any student studying ERP systems or data analytics.

The combination of SQL for querying, Pandas for data manipulation, and Matplotlib for visualization provides a comprehensive, end-to-end view of procurement operations — directly mapping to how enterprise tools like SAP work at their core.

Student Signature

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